Multiple Sequence Alignment

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Introduction

Multiple sequence alignment (MSA) is the precursor to phylogeny, motif discovery, and structural modelling: it arranges residues so homologous positions align in columns. The `msa` Bioconductor package offers ClustalW, ClustalOmega, and Muscle under a unified R interface.

Prerequisites

FASTA sequences; substitution scoring (PAM, BLOSUM for proteins; match/mismatch for DNA).

Theory

Progressive alignment (ClustalW): build a guide tree from pairwise distances, then align sequences following the tree from leaves to root. **Iterative refinement** (Muscle, MAFFT) revisits partial alignments to improve scores. **Consistency-based** methods (T-Coffee) combine information from multiple library alignments.

Accuracy depends on sequence divergence: for highly divergent sequences, structural alignment (DALI for proteins) outperforms sequence-only methods.

Assumptions

Sequences are homologous; a single substitution model approximates the evolutionary process; insertions/deletions are rare enough for gap parameters to be meaningful.

R Implementation

```
library(Biostrings); library(msa)

# Example amino-acid sequences (hemoglobin across species)
aas <- readAAStringSet(
  system.file("examples", "exampleAA.fasta", package = "msa"))
aas
aln <- msa(aas, method = "ClustalW", verbose = FALSE)
aln

# Summary of alignment
print(aln, show = "complete")

# Convert to phangorn for phylogeny
library(phangorn)
aln_phy <- as.phyDat(as(aln, "AAStringSet"), type = "AA")
dm <- dist.ml(aln_phy)
tree <- NJ(dm)
plot(tree, main = "NJ tree from MSA", cex = 0.8)
```

Output & Results

An MSA object with aligned sequences and gap characters; a consensus is derivable via `consensusString`.

Interpretation

"ClustalW alignment of the hemoglobin orthologs reveals conserved haem-binding residues and lineage-specific substitutions on the surface helices."

Practical Tips

- For > 1000 sequences, MAFFT (outside R) is substantially faster than Biostrings/msa.
- Alignment quality matters more than tree-building algorithm; a bad MSA produces a bad phylogeny.
- Trim alignment columns with many gaps (e.g., `trimAl`) before tree inference.
- Visualise alignments with `ggmsa` to identify problematic columns.
- Amino-acid alignments are often more reliable than codon alignments for distantly related sequences; use back-translation when codon-level analysis is needed.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- RNA-seq with DESeq2 / edgeR
- Enrichment analysis (GSEA / ORA)
- scRNA-seq with Seurat